



12397 - Where Are the Stars? A Search for the Stellar Counterparts of Two Off-Nuclear TDE Candidates

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) eRASS_J1421-29_host
	2		NIRCam Imaging	(2) NGC_6099_HLX-1_host

ABSTRACT

Off-nuclear tidal disruption events (TDEs) have recently emerged as powerful probes of intermediate-mass black holes (IMBHs) lurking in compact stellar systems beyond galactic nuclei. Two of the most promising candidates—NGC~6099~HLX-1 and eRASS~J1421–29—exhibit luminous, thermal X-ray emission and characteristic TDE light curves, yet lack any detectable pre-flare stellar counterparts. Their locations, tens of kiloparsecs from massive host galaxies, imply that these IMBHs must reside in extremely faint ultra-compact dwarfs or massive star clusters. We propose deep \textit{JWST}/NIRCam imaging in six filters (F090W/F444W, F115W/F227W, F150W/F356W) to measure the full 1–5~ μ m SEDs of these systems and uncover their underlying stellar populations. These observations will (i) determine the stellar masses, ages, and morphologies of the host systems; (ii) constrain stellar densities and dynamical environments; and (iii) combine with X-ray and optical black hole mass estimates to extend black hole–host scaling relations into the IMBH regime. This program will reveal whether off-nuclear TDEs trace a hidden population of black holes

bridging the gap between stellar-mass and supermassive systems, providing the first direct view of the stellar environments that host accreting IMBHs.

OBSERVING DESCRIPTION

We will obtain deep JWST/NIRCam imaging of NGC 6099 HLX-1 and eRASS J1421–29 using six filters (F090W/F444W, F115W/F227W, F150W/F356W) to measure their full 1–5 μm SEDs and search for faint stellar hosts. Each visit will use both NIRCam modules with standard 4-point dithers and the FASTR1 readout. Exposure times of 25 and 40 minutes per filter pair will achieve $S/N > 5$ for systems as faint as $\sim 5 \times 10^4 M_\odot$, enabling detections or stringent limits on compact hosts. Data will be analyzed with PSF-matched photometry and GALFIT modeling to derive stellar masses, ages, and effective radii, providing a direct test of the intermediate-mass black hole host interpretation and extend black hole host galaxy scalings to the IMBH regime.

Proposal 12397 - Targets - Where Are the Stars? A Search for the Stellar Counterparts of Two Off-Nuclear TDE Candidates

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	eRASS_J1421-29_host	RA: 14 21 39.6240 (215.4151000d) Dec: -29 53 21.12 (-29.88920d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]				
Fixed Targets	(2)	NGC_6099_HLX-1_host	RA: 16 15 34.2960 (243.8929000d) Dec: +19 27 7.92 (19.45220d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]				

Proposal 12397 - Observation 1 - Where Are the Stars? A Search for the Stellar Counterparts of Two Off-Nuclear TDE Candidates

Observation	Proposal 12397, Observation 1										Tue May 26 22:00:50 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	eRASS_J1421-29_host	RA: 14 21 39.6240 (215.4151000d) Dec: -29 53 21.12 (-29.88920d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	SHALLOW2	6	1	8	8	2319.142		
	2	F150W	F356W	SHALLOW2	6	1	8	8	2319.142		
	3	F200W	F444W	SHALLOW2	6	1	8	8	2319.142		
Special Requirements	Offset 20.0 arcsec, 20.0 arcsec										

Proposal 12397 - Observation 2 - Where Are the Stars? A Search for the Stellar Counterparts of Two Off-Nuclear TDE Candidates

Observation	Proposal 12397, Observation 2										Tue May 26 22:00:50 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC_6099_HLX-1_host	RA: 16 15 34.2960 (243.8929000d) Dec: +19 27 7.92 (19.45220d) Equinox: J2000								
	Comments: Category=Galaxy Description=[Compact dwarf galaxy]										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	BRIGHT2	6	1	8	8	1030.73		
	2	F150W	F356W	BRIGHT2	6	1	8	8	1030.73		
	3	F200W	F444W	BRIGHT2	6	1	8	8	1030.73		
Special Requirements	Offset 20.0 arcsec, 20.0 arcsec										